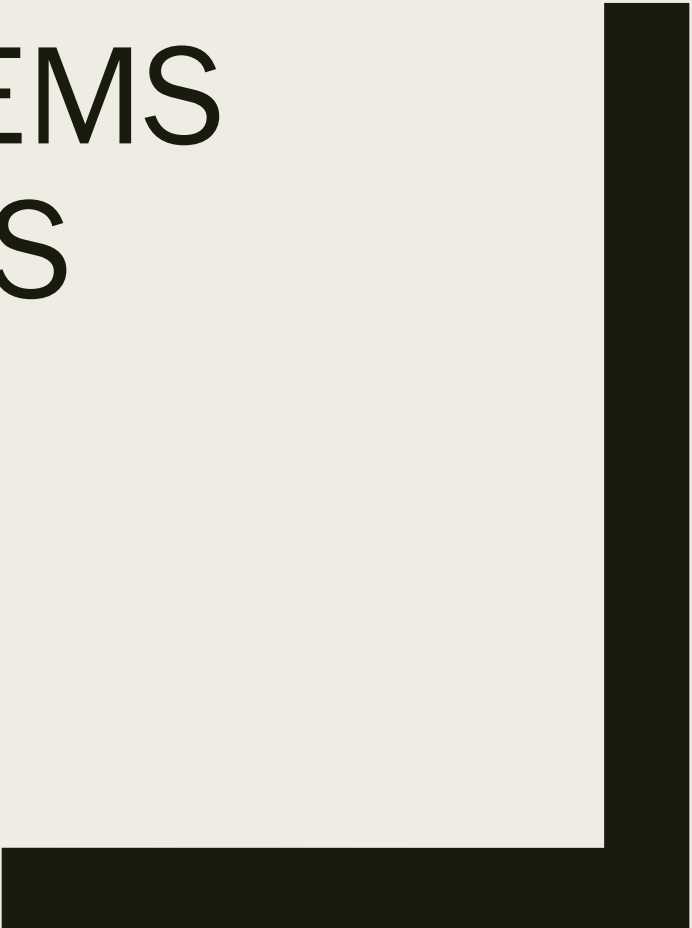




COMPUTER SYSTEMS FUNDAMENTALS (4COSC004W)



In this lecture we will cover:

- Signed integers
 - *Two's Complement*
 - *Ranges of values*
 - *Binary subtraction*

SIGNED INTEGERS

Negative numbers
Two's Complement

By the end of this unit, you will:

- Appreciate the significance of signed values
- Be able to represent signed Binary values using Two's Complement
 - *Appreciate the ranges of signed Binary values*
- Be able to perform Binary subtractions
- Revisit primitive arithmetic functions

Negative numbers

- Consider Decimal
 - *Negative 12*
 - -12
- But no *signage* in Binary
- Two's Complement

Two's Complement

- Representation of Negative Binary numbers
 - We are going to work in an 8-bit system
- Flip bits (NOT)
- Add the value **1**

Two's Complement -15								
15 :					1	1	1	1
NOT	1	1	1	1	0	0	0	0
+1								1
-15 :	1	1	1	1	0	0	0	1

Two's Complement -5								
5 :						1	0	1
NOT	1	1	1	1	1	0	1	0
+1								1
-5 :	1	1	1	1	1	0	1	1

Two's Complement -16								
16 :				1	0	0	0	0
NOT	1	1	1	0	1	1	1	1
+1								1
-16 :	1	1	1	1	0	0	0	0

Unsigned Binary – Base 2

- 8 Bits
 - Values: $0 \rightarrow 255$
 - $0 \rightarrow (2^8 - 1)$
 - 2^8 values
- n Bits
 - Values: $0 \rightarrow (2^n - 1)$
 - 2^n values

Signed Binary – Base 2

- 8 Bits
 - 2^8 values
 - Values: $-128 \rightarrow 0 \rightarrow +127$
 - $-(2^{8-1}) \rightarrow 0 \rightarrow ((2^{8-1}) - 1)$
- n Bits
 - 2^n values
 - Values: $-(2^{n-1}) \rightarrow 0 \rightarrow ((2^{n-1}) - 1)$

$$42 + 13 = 55$$

	Binary addition: 42 + 13								Dec
			1	0	1	0	1	0	42
					1	1	0	1	13
			1	1	0	1	1	1	55
				1					

55 - 13

	Binary representation of 55								Dec
			1	1	0	1	1	1	55

	Two's Complement -13							
13 :					1	1	0	1
NOT	1	1	1	1	0	0	1	0
+1								1
-13 :	1	1	1	1	0	0	1	1

Overflow	Binary addition: 55 + (-13)								Dec
			1	1	0	1	1	1	55
	1	1	1	1	0	0	1	1	-13
	0	0	1	0	1	0	1	0	42
1	1	1	1		1	1	1		

$$55 - 13 = 42$$

Overflow	Binary addition: 55 + (-13)								Dec
			1	1	0	1	1	1	55
	1	1	1	1	0	0	1	1	-13
	0	0	1	0	1	0	1	0	42
1	1	1	1		1	1	1		
1	0	0	1	0	1	0	1	0	298

$$298 - 256 = 42$$

Primitive arithmetic functions performed by the CPU (ALU) - revisited

- Simple, fast Binary operations:
 - *Addition (last week)*
 - *Multiplication by 2 (Left Shift)*
 - *Division by 2 (Right Shift)*
 - *Subtraction (Two's Complement & addition)*

In this lecture we covered:

- Signed integers
 - *Two's complement*
 - *Range of values*
 - *Subtraction of Binary values*

Further reading:

- Computer Science Illuminated
 - *Chapter 3 (p64.5)*
- Foundation Maths
 - *Chapter 14*

Thank you

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